Complex Differential Geometry on the Helical Manifold

Parts I–II: Single Cell & Stacked Chain — Axioms 1–6, Corollaries 1–10

Contraction Integrals Throughout · Leibniz Notation Only

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Prerequisite Reading

This paper uses contraction integrals exclusively—no Riemann sums appear anywhere. The contraction integral is an alternative geometric framework for integration based on level-set contraction and the layer-cake decomposition. Its central tool is **inverse duality**: for a strictly monotone continuous f on $[a, b]$,

$$\int_a^b f + \int_{f(a)}^{f(b)} f^{-1} = b \cdot f(b) - a \cdot f(a).$$

The companion paper, "The Contraction Integral: Integration via Level-Set Contraction" (Yazdani, 2026), develops this framework from first principles and should be read first.

Abstract

Six axioms define a helical 2-manifold embedded as a surface in $\mathbb{R}^3$. From these axioms, together with standard differential geometry (explicitly imported where required), we obtain: an induced metric with $g^{uu} = \frac{1}{2}$ (fixing $c = 1$); a constant gauge connection $A_u = 2\pi k$ whose holonomy identifies k as a topological winding number; a gauge-diagonalised cone metric; geodesic-derived radial dynamics with an effective potential $V(u) = k^2/(2(1+u)^2)$; a proof that local physics is cell-independent; integer-quantised angular momentum from the closure condition; a per-mode radial ODE (Euler's equation on the cone); and a phase-matching condition $\ell = k$ from holonomy coherence.

The radial ODE admits: interpretation as the separated form of a wave equation on the manifold; a scalar first-order factorisation analogous to relativistic factorisation procedures; and reduction to a Schrödinger-type normal form.

The Laplace–Beltrami operator and wave equation $\Box_{\mathcal{M}}\psi = 0$ are **imported from standard Riemannian geometry** and are not derived from the axioms. Any physical interpretation (Klein–Gordon, Dirac, Schrödinger analogies) is presented as **structural correspondence**, not equivalence.

All integrals are contraction integrals computed via inverse duality. All derivatives in Leibniz notation. One manifold. One parameter.

1 The Six Axioms

Axiom 1 (Time is Self-Referential)**.**

$$\frac{dt}{dt} = 1.$$

This fixes $c = 1$ in natural units. The z-axis grows as $z = 1 + u$, with $u \in [0, 1]$ the affine parameter within one cell.

Axiom 2 (Coupled Radius)**.** *Radius is derived from the angular variable ($r = \theta/\pi$) and the spiral expands as a light cone: $dr/dz = 1$. For one cell:*

$$r(u) = 1 + u.$$

Axiom 3 (Light-Cone Expansion)**.** *The radial expansion rate equals the speed of light: $dr/dz = 1$. Combined with Axiom 2, this couples the transverse radius to the longitudinal coordinate.*

Axiom 4 (Helical Motion)**.**

$$\theta(u, v) = v + 2\pi k\, u, \qquad k \in \mathbb{Z}^+.$$

The winding number k is the single free geometric parameter.

Axiom 5 (Reality Embedding)**.**

$$\mathbf{R}(u, v) = \big((1 + u)\cos\theta,\ (1 + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k\, u.$$

Axiom 6 (Unitary Closure)**.** *The spiral closes at integer z:*

$$\psi(u, v + 2\pi) = \psi(u, v).$$

This forces every transverse mode to carry integer angular momentum.

2 Corollaries

Corollary 1 (Induced Metric)**.** *The first fundamental form on $\mathcal{M}$ is computed by taking partial derivatives of the embedding $\mathbf{R}(u, v)$ with respect to u and v, then forming dot products.*

The tangent vectors are:

$$\mathbf{R}_u = \frac{\partial \mathbf{R}}{\partial u}, \qquad \mathbf{R}_v = \frac{\partial \mathbf{R}}{\partial v}.$$

The metric components $E = \mathbf{R}_u \cdot \mathbf{R}_u$, $F = \mathbf{R}_u \cdot \mathbf{R}_v$, $G = \mathbf{R}_v \cdot \mathbf{R}_v$ evaluate to:

$$E = 4\pi^2 k^2 (1 + u)^2 + 2, \tag{1}$$

$$F = 2\pi k\,(1 + u)^2, \tag{2}$$

$$G = (1 + u)^2. \tag{3}$$

The metric determinant and its square root:

$$\det(g) = EG - F^2 = 2(1 + u)^2, \qquad \sqrt{g} = \sqrt{2}\,(1 + u).$$

The inverse metric component:

$$g^{uu} = \frac{G}{\det(g)} = \frac{(1 + u)^2}{2(1 + u)^2} = \frac{1}{2}.$$

Derivation. *The embedding gives:*

$$\mathbf{R}_u = \big(\cos\theta - (1+u)(2\pi k)\sin\theta,\ \ \sin\theta + (1+u)(2\pi k)\cos\theta,\ \ 1\big),$$
$$\mathbf{R}_v = \big(-(1+u)\sin\theta,\ \ (1+u)\cos\theta,\ \ 0\big).$$

Computing dot products and applying $\sin^2\theta + \cos^2\theta = 1$*:*

$$\begin{aligned} E &= \cos^2\theta + (1+u)^2(2\pi k)^2\sin^2\theta - 2(1+u)(2\pi k)\cos\theta\sin\theta \\ &\quad + \sin^2\theta + (1+u)^2(2\pi k)^2\cos^2\theta + 2(1+u)(2\pi k)\sin\theta\cos\theta + 1 \\ &= 1 + 4\pi^2k^2(1+u)^2 + 1 = 4\pi^2k^2(1+u)^2 + 2. \end{aligned}$$

The cross terms in $\sin\theta\cos\theta$ *cancel. Similarly, the cross terms cancel in* F *and* G*, yielding the stated results. The determinant follows from direct computation:*

$$EG - F^2 = \big(4\pi^2k^2(1+u)^2 + 2\big)(1+u)^2 - 4\pi^2k^2(1+u)^4 = 2(1+u)^2.$$

Verified symbolically via SymPy.

Corollary 2 (Holonomy — Winding Number as Topological Charge)**.** *The off-diagonal metric component divided by the transverse metric component defines a gauge connection for the* v*-fibre:*

$$A_u = \frac{F}{G} = \frac{2\pi k\,(1+u)^2}{(1+u)^2} = 2\pi k.$$

Interpretation. A_u *measures how much phase (the* v*-direction) gets dragged per step in* u*. Because both* F *and* G *scale as* $(1+u)^2$*, their ratio is constant: the helix winds at a uniform rate regardless of where you are in the cell.*

The holonomy over one cell:

$$\Phi_{\text{cell}} = \int_0^1 A_u\,du = \int_0^1 2\pi k\,du = 2\pi k.$$

This is exact: k *complete phase cycles per cell. Since* $k \in \mathbb{Z}^+$*, the phase returns to itself* ($e^{i\cdot 2\pi k} = 1$)*, making* k *a* ***topological invariant****—it cannot be changed by continuous deformation of the manifold.*

The holonomy selects k *as the fundamental geometric quantum number of the helix.*

Verified symbolically via SymPy.

Corollary 3 (Christoffel Symbols)**.** *Computed from the standard formula*

$$\Gamma^i_{jl} = \frac{1}{2}\sum_m g^{im}\left(\frac{\partial g_{mj}}{\partial x^l} + \frac{\partial g_{ml}}{\partial x^j} - \frac{\partial g_{jl}}{\partial x^m}\right),$$

using the metric (1)–(3) *and its inverse. The six independent symbols are:*

$$\Gamma^u_{uu} = -2\pi^2k^2(u+1), \qquad \Gamma^v_{uu} = \frac{4\pi k\big(\pi^2k^2u^2 + 2\pi^2k^2u + \pi^2k^2 + 1\big)}{u+1}, \tag{4}$$

$$\Gamma^u_{uv} = -\pi k\,(u+1), \qquad \Gamma^v_{uv} = \frac{2\pi^2k^2u^2 + 4\pi^2k^2u + 2\pi^2k^2 + 1}{u+1}, \tag{5}$$

$$\Gamma^u_{vv} = -\frac{u+1}{2}, \qquad \Gamma^v_{vv} = \pi k\,(u+1). \tag{6}$$

Verified symbolically via SymPy and cross-checked numerically at $u = 1/2$***,*** $k = 1$***.***

Corollary 4 (Cell Action via Contraction)**.** *The action (surface area) of one cell:*

$$S = \int_0^{2\pi} \int_0^1 \sqrt{g}\, du\, dv = 2\pi \int_0^1 \sqrt{2}\,(1+u)\, du.$$

Set $h(u) = \sqrt{2}\,(1+u)$. Then $h(0) = \sqrt{2}$, $h(1) = 2\sqrt{2}$, and $h^{-1}(t) = t/\sqrt{2} - 1$.
By inverse duality:

$$\int_0^1 h(u)\, du = 1 \cdot h(1) - 0 \cdot h(0) - \int_{\sqrt{2}}^{2\sqrt{2}} h^{-1}(t)\, dt = 2\sqrt{2} - \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}.$$

Therefore:

$$S = 2\pi \times \frac{3\sqrt{2}}{2} = 3\sqrt{2}\,\pi.$$

The action is ***independent of*** *k. The winding number does not affect the surface area of the cell.*

Verified symbolically via SymPy.

Corollary 5 (Phase Energy via Geodesic Derivation)**.** ***Origin of the potential.*** *The coordinate change $v' = v + 2\pi k u$ removes the off-diagonal metric term, yielding the cone metric:*

$$ds^2 = 2\, du^2 + (1+u)^2\, dv'^2.$$

The geodesic equations on this cone are:

$$\frac{d^2u}{ds^2} - \frac{(1+u)}{2}\left(\frac{dv'}{ds}\right)^2 = 0,$$
$$\frac{d}{ds}\left[(1+u)^2 \frac{dv'}{ds}\right] = 0.$$

The second equation gives conservation of angular momentum: $(1+u)^2\, dv'/ds = L = const.
Substituting $dv'/ds = L/(1+u)^2$ into the first equation:

$$\frac{d^2u}{ds^2} = \frac{L^2}{2(1+u)^3}.$$

Reducing to first order via $p\, dp/du = d^2u/ds^2$ and integrating:

$$p^2 = E_{total} - \frac{L^2}{2(1+u)^2}.$$

From the metric normalisation $2p^2 + L^2/(1+u)^2 = 1$, the total energy is $E_{total} = 1/2 = g^{uu}$.
The position-dependent term is the geometric potential:

$$V(u) = \frac{k^2}{2(1+u)^2},$$

where we identify $L = k$ (the angular momentum equals the winding number).

Integration. *The integrated potential per cell:*

$$E_{phase} = \int_0^1 \frac{k^2}{2(1+u)^2}\, du.$$

By the layer-cake decomposition, the level width of $V(u) = k^2/(2(1+u)^2)$ (strictly decreasing on $[0,1]$) is:

$$w_V(t) = \begin{cases} 1 & 0 \le t < k^2/8, \\ \dfrac{k}{\sqrt{2t}} - 1 & k^2/8 \le t \le k^2/2. \end{cases}$$

Contraction integral:

$$\int_0^{k^2/2} w_V(t)\,dt = \frac{k^2}{8} + \frac{k^2}{8} = \frac{k^2}{4}.$$

$$\boxed{E_{phase} = \frac{k^2}{4}.}$$

Consistency check: *Integrated kinetic energy + integrated potential* $= (1/2 - k^2/4) + k^2/4 = 1/2 = g^{uu}$. ✓

Verified symbolically via SymPy.

Corollary 6 (Cell-Independence of the Local Metric)**.** ***Statement.*** *The quantities governing local physics on each cell are independent of the cell number n.*

Proof. *Parameterise cell n with base radius a (where $a = n + 1$ for cell n in the global chain), so the embedding becomes:*

$$\mathbf{R}_a(u, v) = \big((a + u)\cos\theta,\ (a + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k u.$$

The metric components on this general cell are:

$$\begin{aligned} E(a) &= 4\pi^2 k^2 (a + u)^2 + 2, \\ F(a) &= 2\pi k\,(a + u)^2, \\ G(a) &= (a + u)^2. \end{aligned}$$

These do *depend on a. However, the physically governing quantities do not:*

1. Gauge connection:

$$A_u = \frac{F(a)}{G(a)} = \frac{2\pi k\,(a + u)^2}{(a + u)^2} = 2\pi k. \qquad \textit{(independent of } a\textit{)}$$

2. Inverse metric:

$$g^{uu} = \frac{G(a)}{\det g(a)} = \frac{(a + u)^2}{2(a + u)^2} = \frac{1}{2}. \qquad \textit{(independent of } a\textit{)}$$

3. Gauge-diagonalised metric: *The coordinate change $v' = v + 2\pi k u$ gives*

$$ds^2 = 2\,du^2 + (a + u)^2\,dv'^2$$

on every cell. The wave equation $\Box\psi = 0$ with $\psi = F(u)\,e^{i\ell v'}$ yields Euler's ODE:

$$(a + u)^2 F'' + (a + u)F' - 2\ell^2 F = 0,$$

whose indicial roots $r = \pm\ell\sqrt{2}$ are ***independent of*** *a.*

What does depend on *a:* *The ratio $r(1)/r(0) = (a + 1)/a$, which equals 2 only when $a = 1$. The transfer matrix trace $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$ uses $\ln(w(1)/w(0)) = \ln 2$, which requires $a = 1$.*

Conclusion. *The chain is periodic: each cell resets to $a = 1$, with $u \in [0, 1]$ as a local coordinate. The gauge connection, the inverse metric, and the indicial structure of the wave equation are cell-independent. The transfer matrix (derived in later corollaries) requires this periodicity to achieve $\det T = 1/2$ and $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$.*

Verified symbolically via SymPy.

3 Part II: The Stacked Chain

Corollary 7 (Fourier Decomposition on the Transverse Circle)**.** *Axiom 6 requires $\psi(u, v+2\pi) = \psi(u, v)$. A function $e^{i\ell v}$ satisfies this if and only if:*

$$e^{i\ell(v+2\pi)} = e^{i\ell v} \quad \Longleftrightarrow \quad e^{2\pi i\ell} = 1 \quad \Longleftrightarrow \quad \ell \in \mathbb{Z}.$$

Non-integer values fail: for example, $\ell = 1/2$ gives $e^{i\pi} = -1$, which would require $\psi(u, v + 2\pi) = -\psi(u, v)$—a weaker closure condition not permitted by Axiom 6.

The general solution on cell n is therefore:

$$\psi_n(u, v) = \sum_{\ell=-\infty}^{\infty} c_\ell^{(n)} f_\ell^{(n)}(u)\, e^{i\ell v}, \qquad \ell \in \mathbb{Z}.$$

No truncation. The full tower of integer angular momentum modes is present. The quantisation of ℓ is not imposed—it is forced by the closure condition.

Verified symbolically via SymPy *(integer ℓ gives $e^{2\pi i\ell} = 1$; half-integer gives -1).*

Corollary 8 (Per-Mode Radial ODE)**.** ***Imported machinery.*** *The Laplace–Beltrami operator is the unique second-order differential operator on a Riemannian manifold that is (a) coordinate-invariant, (b) self-adjoint, and (c) reduces to the ordinary Laplacian on flat space. It is* not derived from the axioms. *We apply known Riemannian geometry to the metric our axioms produced:*

$$\Box_{\mathcal{M}}\psi = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i}\left(\sqrt{g}\, g^{ij} \frac{\partial \psi}{\partial x^j}\right).$$

Two correction factors versus the flat Laplacian: g^{ij} accounts for non-orthogonal coordinate directions; $\sqrt{g}$ accounts for varying patch area across the surface.

In the original *(u, v)* ***frame.*** *Substituting $\psi = f(u)\, e^{i\ell v}$ into $\Box_{\mathcal{M}}\psi = 0$ and dividing by $e^{i\ell v}$ yields a second-order ODE for $f(u)$ that depends on both k (from the metric) and ℓ (from the Fourier mode). The full expression is large due to the off-diagonal metric coupling.*

In the gauge-diagonalised frame. *The coordinate change $v' = v+2\pi ku$ reduces the metric to the cone $ds^2 = 2\, du^2 + (1+u)^2\, dv'^2$. The same Laplace–Beltrami operator with $\psi = F(u)\, e^{i\ell v'}$ yields Euler's ODE:*

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0.$$

The indicial equation $r^2 = 2\ell^2$ gives roots $r = \pm\ell\sqrt{2}$, and the two linearly independent solutions are:

$$F_1(u) = (1+u)^{\ell\sqrt{2}}, \qquad F_2(u) = (1+u)^{-\ell\sqrt{2}}.$$

Verified symbolically via SymPy: *both solutions give zero residual when substituted back into the ODE.*

Corollary 9 (Phase-Matching Condition — Mode $\ell = k$)**.** *Each Fourier mode $e^{i\ell v}$ carries angular momentum ℓ. The gauge connection (Corollary 2) contributes holonomy $2\pi k$ per cell. The relative phase mismatch per cell is:*

$$\delta\phi_\ell = 2\pi(\ell - k).$$

Resonant mode *($\ell = k$): $\delta\phi_k = 0$. The mode propagates coherently across cells with no phase drift.*

Non-resonant modes *($\ell \neq k$): Over N cells the accumulated phase is $2\pi N(\ell - k)$. For $N \gg 1$, these contributions destructively interfere and the mode decays.*

This identifies $\ell = k$ as a ***natural phase-matching condition****: the mode whose angular momentum matches the winding number is the one selected by coherence.*

Caveat. *This condition arises from phase alignment, not from a derived dynamical suppression law. No claim is made here that non-resonant modes are forbidden—only that they fail to propagate coherently across many cells. A rigorous decay estimate (e.g. bounding $\|T_\ell^N\|$ for $\ell \neq k$) would strengthen this to a derived selection rule, and is deferred to later work.*

Verified symbolically via SymPy.

Corollary 10 (Structural Correspondences: Wave Equation, Factorisation, and Normal Form)**.** *The Euler ODE from Corollary 8,*

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0,$$

admits three algebraic reductions. Each exhibits structural correspondence with a known equation of physics. No ansatz is used—only operator algebra and change of dependent variable. The correspondences are presented as ***structural analogies, not claimed equivalences****.*

1. Wave equation structure (Klein–Gordon analogy). *The Euler ODE arises from $\Box_{\mathcal{M}}\psi = 0$ after separation of variables. It is the full second-order wave equation on the manifold. This provides a Klein–Gordon-type structure, but relies on the* imported *Laplace–Beltrami operator (Corollary 8).*

2. Scalar first-order factorisation (Dirac analogy). *Define the Euler operator $D = (1+u)\, d/du$. Then:*

$$D^2F = (1+u)\frac{d}{du}\Big[(1+u)F'\Big] = (1+u)^2F'' + (1+u)F'.$$

So the Euler ODE is $D^2F - 2\ell^2F = 0$, which factors algebraically:

$$\big(D - \ell\sqrt{2}\big)\big(D + \ell\sqrt{2}\big)F = 0.$$

Each factor is a first-order equation:

$$(1+u)\, F' + \ell\sqrt{2}\, F = 0,$$
$$(1+u)\, F' - \ell\sqrt{2}\, F = 0.$$

Each reproduces one of the two indicial solutions:

$$F = (1+u)^{-\ell\sqrt{2}} \quad \textit{and} \quad F = (1+u)^{+\ell\sqrt{2}}$$

respectively. The factorisation is verified by checking that both residuals vanish.

Caveat. *This is a* scalar *first-order factorisation. It is structurally analogous to Dirac's procedure of factoring a second-order operator into first-order pieces, but it does not introduce spinors, Clifford algebra, or additional internal degrees of freedom. The analogy is algebraic, not physical, until further structure is derived.*

3. Schrödinger-type normal form. *To eliminate the first-derivative term from the Euler ODE algebraically, substitute $F = (1+u)^\alpha\, h(u)$ and choose α to kill the h' coefficient. The requirement is $2\alpha + 1 = 0$, giving $\alpha = -1/2$. With $F = (1+u)^{-1/2}\, h(u)$:*

$$(1+u)^2\, h'' + \big(\tfrac{1}{4} - 2\ell^2\big)\, h = 0.$$

No h' term remains. Dividing by $-2(1+u)^2$:

$$-\frac{1}{2}\, h'' + \frac{2\ell^2 - \frac{1}{4}}{2(1+u)^2}\, h = 0.$$

This is a Schrödinger-type normal form (Sturm–Liouville structure): $-\frac{1}{2}h'' + V_{\text{eff}}(u)\,h = 0$ with effective potential $V_{\text{eff}} = (2\ell^2 - 1/4)/(2(1+u)^2)$, which for $\ell \gg 1$ approaches the geodesic potential $\ell^2/(1+u)^2$ from Corollary 5.

Caveat. *This is a* time-independent *normal form. It is not yet a full time-dependent Schrödinger equation; the role of u as an evolution parameter is not formally established within the present axioms.*

4. Coordinate sensitivity: origin of i. *In the gauge-diagonalised cone frame, the ODE is purely real. In the* original (u, v) *frame, the inverse metric has $g^{uv} = -\pi k$, and the cross-term $g^{uv}\,\partial_u\partial_v\psi$ introduces imaginary contributions. On the mass shell ($\ell = k$), the coefficient of $f'(u)$ separates as:*

$$\textit{coeff}(f') = \underbrace{\frac{1}{2(1+u)}}_{\textit{real: geometric damping}} - \underbrace{2i\pi k^2}_{\textit{imaginary: constant}}.$$

The imaginary part $-2i\pi k^2$ is constant because $g^{uv} = -\pi k$ is constant, and it vanishes in the cone frame because the gauge diagonalisation removes g^{uv} entirely.

Caveat. *This imaginary structure is* coordinate-dependent*: it is present in the (u, v) frame and absent in the (u, v') frame. Its physical significance (if any) remains to be established. Both frames describe the same geometry; the i is a feature of the chart, not yet shown to be a feature of the physics.*

Summary. *Three levels of the same equation:*

Structure	***Operation***	***Result***	***Status***
Wave eqn	*Full Euler ODE*	$(1+u)^2F'' + (1+u)F' - 2\ell^2F = 0$	*Imported* $\Box_{\mathcal{M}}$
Factorisation	$(D-r)(D+r)F = 0$	*Two first-order equations*	*Scalar only*
Normal form	$F = (1+u)^{-1/2}h$	$-\frac{1}{2}h'' + V_{\text{eff}}\,h = 0$	*Time-indep.*

Verified symbolically via SymPy: *operator factorisation residuals zero; normal-form reduction confirmed; f' coefficient splits into real and imaginary parts as stated.*

4 Interpretation Boundary

The present work distinguishes between three categories of result:

Derived from axioms and standard differential geometry:

Metric · Connection · Holonomy · Christoffel symbols · Geodesics · Gaussian curvature · Cell action · Cell-independence · Fourier quantisation · Phase-matching condition

Imported from standard Riemannian geometry:

Laplace–Beltrami operator · Wave equation $\Box_{\mathcal{M}}\psi = 0$

Interpretive (structural correspondence, not equivalence):

Klein–Gordon analogy · Dirac-like factorisation · Schrödinger normal form · Phase-matching as "mass shell" · Imaginary coefficient as physical i

These correspondences are suggestive. Whether any of them can be promoted to physical equivalences requires: (a) a derived identification of u with a physical evolution parameter; (b) an observer or measurement framework; and (c) predictions that can be tested against experiment.

Part III: Inter-Cell Matching and Spectrum

Corollary 11 (Inter-Cell Transfer Matrix). At cell junctions, the wave equation $\Box_M \psi = 0$ requires continuity of both the mode function and its derivative:

$$f_\ell^{(n)}(1) = f_\ell^{(n+1)}(0), \qquad \left.\frac{df_\ell^{(n)}}{du}\right|_{u=1} = \left.\frac{df_\ell^{(n+1)}}{du}\right|_{u=0}.$$

Why C^1. The ODE is second-order. A discontinuity in f' at a junction would produce a delta function in f'', requiring a point source at the boundary. Since no sources are postulated, both f and f' must be continuous. Two matching conditions at each junction, two linearly independent solutions per cell—the propagation is completely determined by a 2×2 matrix.

Construction. From Corollary 8, the gauge-diagonalised cone metric yields Euler's equation with solutions $F_1 = (1+u)^p$ and $F_2 = (1+u)^{-p}$, where $p = \ell\sqrt{2}$. Their values and derivatives at the cell boundaries are:

$$F_1(0) = 1, \quad F_1'(0) = p, \quad F_1(1) = 2^p, \quad F_1'(1) = p \cdot 2^{p-1},$$

$$F_2(0) = 1, \quad F_2'(0) = -p, \quad F_2(1) = 2^{-p}, \quad F_2'(1) = -p \cdot 2^{-p-1}.$$

The transfer matrix is built by propagating two canonical initial conditions:

Column 1 ($f(0) = 1,\ f'(0) = 0$): $A = B = 1/2$.

$$f(1) = \frac{2^p + 2^{-p}}{2} = \cosh(p \ln 2), \qquad f'(1) = \frac{p}{2}\sinh(p \ln 2).$$

Column 2 ($f(0) = 0,\ f'(0) = 1$): $A = 1/(2p),\ B = -1/(2p)$.

$$f(1) = \frac{\sinh(p \ln 2)}{p}, \qquad f'(1) = \frac{1}{2}\cosh(p \ln 2).$$

Therefore:

$$T_\ell = \begin{pmatrix} \cosh(p \ln 2) & \dfrac{\sinh(p \ln 2)}{p} \\ \dfrac{p}{2}\sinh(p \ln 2) & \dfrac{1}{2}\cosh(p \ln 2) \end{pmatrix}, \qquad p = \ell\sqrt{2}.$$

Immediate properties.

$$\operatorname{tr} T_\ell = \cosh(p \ln 2) + \tfrac{1}{2}\cosh(p \ln 2) = \tfrac{3}{2}\cosh(\ell\sqrt{2}\,\ln 2).$$

$$\det T_\ell = \tfrac{1}{2}\cosh^2(p \ln 2) - \tfrac{1}{2}\sinh^2(p \ln 2) = \tfrac{1}{2}.$$

The determinant is not unity—it equals $\sqrt{g(0)}/\sqrt{g(1)} = \sqrt{2}\,/\,(2\sqrt{2}) = 1/2$, recording the metric expansion from cell entry to cell exit (Corollary 13 derives this from the Wronskian).

Degenerate case ($\ell = 0$): The double root $p = 0$ gives $F_1 = 1$, $F_2 = \ln(1+u)$, and:

$$T_0 = \begin{pmatrix} 1 & \ln 2 \\ 0 & 1/2 \end{pmatrix}.$$

Propagation. By Corollary 6, each cell has the same local metric, hence the same T_ℓ. After N cells:

$$\begin{pmatrix} f(N) \\ f'(N) \end{pmatrix} = T_\ell^N \begin{pmatrix} f(0) \\ f'(0) \end{pmatrix}.$$

Riccati perspective. The substitution $z = F'/F$ converts the Euler ODE to the Riccati equation:

$$\frac{dz}{du} = -z^2 - \frac{z}{1+u} + \frac{2\ell^2}{(1+u)^2}.$$

Under the log coordinate $s = \ln(1+u)$ and the rescaling $Z = (1+u)\,z$, this autonomises to:

$$\frac{dZ}{ds} = -Z^2 + 2\ell^2,$$

with constant fixed points $Z_\pm = \pm\ell\sqrt{2}$. Both fixed points live in the same coordinate—there is no regime transition, no coordinate incompatibility, no special function. This is the trivial case of Riccati flow: WKB is exact on the cone because the Euler equation has constant coefficients in log coordinates.

The two columns of T_ℓ correspond to the two fixed-point trajectories: each solution $F = (1+u)^{\pm p}$ follows a single Riccati fixed point from $u = 0$ to $u = 1$ without deviation.

Connection to the special functions framework. The companion paper "What Is So Special About 'Special Functions'?" (Yazdani, 2026) argues that every classical special function is a smooth Riccati flow trajectory connecting two asymptotic regimes with incompatible autonomising coordinates. The Euler equation on the cone is the degenerate case: only one autonomising coordinate ($\ln w$) is needed, both regimes are compatible, and the flow reduces to constant fixed points. No scar forms. The transfer matrix T_ℓ is the finite propagator of this trivial flow.

Verification.

- *Symbolic*: Residuals of both solutions $F = (1+u)^{\pm\ell\sqrt{2}}$ vanish identically when substituted into the Euler ODE (SymPy).
- *Numeric spot checks*: Residuals exactly 0 across $\ell \in \{1, 2, 3, 5, 50\}$, $u \in \{1, 2, 3, 5, 50\}$ (SymPy, exact rational arithmetic).
- *Riccati solver*: Autonomous classification recovered; both fixed points identified; no poles on $[0, 1]$.
- *Numerical integration*: Solution values, transfer matrix entries, and Riccati flow confirmed to machine precision (scipy, errors $< 10^{-11}$).
- *Transfer matrix*: $\det T_\ell = 1/2$ and $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\,\ln 2)$ confirmed for $\ell = 0, 1, 2, 3, 5$ (scipy, errors $< 10^{-10}$).

Corollary 12 (Holonomy Coherence). *To be promoted to revised standard. Sketch: Corollary 9 establishes the phase-matching condition $\ell = k$. This corollary formalises the coherence argument: on the mass shell, $\delta\phi_k = 0$ and modes propagate without phase drift; off-shell, accumulated phase $2\pi N(\ell - k)$ produces destructive interference for $N \gg 1$.*

Corollary 13 (Wronskian Conservation and $\det T = \frac{1}{2}$). *To be promoted to revised standard. Sketch: The Wronskian $W(u) = \sqrt{g(u)}\,(f_1 f_2' - f_2 f_1')$ satisfies $dW/du = 0$. Since $\sqrt{g(0)} = \sqrt{2}$ and $\sqrt{g(1)} = 2\sqrt{2}$, the transfer matrix determinant is $\det T_\ell = \sqrt{g(0)}/\sqrt{g(1)} = 1/2$.*

Corollary 14 (Dispersion Relation). *To be promoted to revised standard. Sketch: Eigenvalues of T_ℓ are $\lambda_\pm = \frac{1}{\sqrt{2}}\, e^{\pm i\kappa}$. Bloch phase: $\cos\kappa = \frac{1}{\sqrt{2}} \operatorname{tr} T_\ell$. Propagating modes: $|\operatorname{tr} T_\ell| \le \sqrt{2}$. Evanescent modes: $|\operatorname{tr} T_\ell| > \sqrt{2}$.*

Corollary 15 (Chain Phase, Action, and Energy). *To be promoted to revised standard. Sketch: Over an N-cell chain, $\Phi_{\text{total}} = 2\pi k N$, $S_{\text{total}} = 3\sqrt{2}\,\pi N$, $E_{\text{total}} = (k^2/4)\,N$. All extensive in N.*

Part IV: Intrinsic Flatness and Field Equations

Corollary 16 (Gaussian Curvature). *To be promoted to revised standard. Sketch: $K(u) = 0$ for all $u \in [0,1]$ and all k. The manifold is intrinsically flat—a developable surface. All curvature experienced by modes is extrinsic.*

Corollary 17 (Extrinsic Curvature). *To be promoted to revised standard. Sketch: Second fundamental form coefficients L, M, N computed from the unit normal. $LN - M^2 = 0$ confirms $K = 0$ (rank-1, developable). Mean curvature $H = \sqrt{2}\,/\,(4(1+u))$.*

Corollary 18 (Gauss–Bonnet and Topology). *To be promoted to revised standard. Sketch: $K = 0$ identically $\Rightarrow \int K\, dA = 0$. Each cell is an annulus ($\chi = 0$). The topology is trivial.*

Corollary 19 (Maxwell's Equations from Gauge Coupling).

Statement. In the gauge-diagonalised cone frame, the wave equation $\Box_M \psi = 0$ is purely real: the Euler ODE for $F(u)$ has no coupling between real and imaginary parts. In the original (u, v) frame, the off-diagonal inverse metric $g^{uv} = -\pi k$ reintroduces the gauge connection, and the radial ODE acquires complex coefficients. Separating $f = a + ib$ (where $a = \operatorname{Re}(f)$, $b = \operatorname{Im}(f)$) yields a pair of coupled real equations whose cross-coupling has the antisymmetric structure of Maxwell's equations.

Derivation. The Laplace–Beltrami operator in the original (u, v) frame, applied to $\psi = f(u)\, e^{i\ell v}$ and divided by $e^{i\ell v}$, gives:

$$\frac{1}{\sqrt{g}}\, \partial_i\big(\sqrt{g}\, g^{ij}\, \partial_j\, \psi\big) \;\longrightarrow\; C_2\, f'' + C_1\, f' + C_0\, f = 0,$$

where the coefficients are computed from the four flux contributions ($g^{uu}\partial_u\partial_u$, $g^{uv}\partial_u\partial_v$, $g^{vu}\partial_v\partial_u$, $g^{vv}\partial_v\partial_v$) with each ∂_v replaced by $i\ell$. After clearing the denominator $(1+u)^2$:

$$C_2 = \tfrac{1}{2}(1+u)^2, \tag{1}$$

$$C_1 = \tfrac{1}{2}(1+u) - 2i\pi\ell k\,(1+u)^2, \tag{2}$$

$$C_0 = \ell\big(-2\pi^2\ell k^2(1+u)^2 - \ell - i\pi k(1+u)\big)\,. \tag{3}$$

On the **mass shell** $\ell = k$ (Corollary 9), these become:

$$C_2 = \tfrac{1}{2}(1+u)^2, \tag{real}$$

$$C_1 = \underbrace{\tfrac{1}{2}(1+u)}_{\text{real: damping}} - \underbrace{2i\pi k^2(1+u)^2}_{\text{imaginary: curl coupling}}, \tag{complex}$$

$$C_0 = \underbrace{-2\pi^2k^4(1+u)^2 - k^2}_{\text{real: potential}} - \underbrace{i\pi k^2(1+u)}_{\text{imaginary: source coupling}}. \tag{complex}$$

Dividing through by $C_2 = \frac{1}{2}(1+u)^2$ gives the **normalised ODE**:

$$f'' + \left(\frac{1}{1+u} - 4i\pi k^2\right) f' + \left(-V - \frac{2i\pi k^2}{1+u}\right) f = 0,$$

where $V = (4\pi^2k^4(1+u)^2 + 2k^2)/(1+u)^2$ is the effective potential (real).

Define:

$$\gamma = 4\pi k^2 \quad \text{(constant)}, \qquad \sigma = \frac{2\pi k^2}{1+u} \quad \text{(position-dependent)}.$$

The coupled real system. Substituting $f = a(u) + i\,b(u)$ and separating real and imaginary parts, the single complex ODE splits into:

$$\begin{aligned} &\textbf{Real equation:} && L[a] &&= -\gamma\, b' \;-\; \sigma\, b, \\ &\textbf{Imaginary equation:} && L[b] &&= +\gamma\, a' \;+\; \sigma\, a, \end{aligned}$$

where $L[\,\cdot\,] = (\,\cdot\,)'' + \frac{1}{1+u}(\,\cdot\,)' - V(\,\cdot\,)$ is the scalar wave operator, identical for both components. The cross-coupling terms are:

Coupling	**In $L[a]$ equation**	**In $L[b]$ equation**	**Structure**
First-order (curl)	$-\gamma\, b' = -4\pi k^2\, b'$	$+\gamma\, a' = +4\pi k^2\, a'$	Antisymmetric, constant
Zeroth-order (source)	$-\sigma\, b = -\frac{2\pi k^2}{1+u}\, b$	$+\sigma\, a = +\frac{2\pi k^2}{1+u}\, a$	Antisymmetric, decays

Both couplings are **antisymmetric**: the coefficient of b' in the real equation is the negative of the coefficient of a' in the imaginary equation, and likewise for b and a. This is the defining algebraic signature of electromagnetism.

Recovery of standard Maxwell. In the **flat limit** (cone corrections removed: $1/(1+u) \to 0$, $V \to \text{const}$, $\sigma \to 0$), the coupled system reduces to:

$$a'' = -\gamma\, b', \qquad b'' = +\gamma\, a'.$$

Integrating once in u (constant of integration zero for wave solutions):

$$\frac{da}{du} = -\gamma\, b = -4\pi k^2\, b, \tag{4}$$

$$\frac{db}{du} = +\gamma\, a = +4\pi k^2\, a. \tag{5}$$

Under the identification $a \leftrightarrow E$ (electric field), $b \leftrightarrow B$ (magnetic field), $u \leftrightarrow z$ (spatial coordinate), these are Maxwell's curl equations in vacuum for a monochromatic plane wave propagating in the z-direction:

$$\frac{dE}{dz} = -\omega\, B, \qquad \frac{dB}{dz} = +\omega\, E, \qquad \omega = 4\pi k^2.$$

Differentiating the first and substituting the second gives the wave equation:

$$\frac{d^2E}{dz^2} + \omega^2\, E = 0.$$

Phase velocity: $v = \omega/\kappa = 1 = c$, by Axiom 1.

The Riemann–Silberstein vector. The complex field $f = a + ib$ is the Riemann–Silberstein vector $\mathbf{F} = \mathbf{E} + i\mathbf{B}$, introduced by Silberstein in 1907 as a reformulation of Maxwell's equations into a single complex equation. The present derivation inverts this: the helix produces f as a single object; E and B are its real and imaginary projections. The Riemann–Silberstein vector is not a mathematical convenience—it is the natural field.

Geometric origin. The cross-coupling exists because $g^{uv} = -\pi k \neq 0$ in the original frame. In the gauge-diagonalised frame ($v' = v + 2\pi k u$), $g^{uv} = 0$ and the ODE is purely real—no coupling, no electromagnetism. The gauge transformation that removes A_u simultaneously removes the electromagnetic structure.

The three corrections that distinguish Maxwell-on-the-cone from flat-space Maxwell are:

1. **Geometric damping**: $P = 1/(1+u)$. Both E and B have a first-derivative self-term encoding $1/r$ amplitude decay from the expanding cone.
2. **Effective potential**: $V(u)$. Second-order self-coupling containing the centrifugal term $k^2/(1+u)^2$ and the gauge-squared term $4\pi^2 k^4$.
3. **Geometric source**: $\sigma = 2\pi k^2/(1+u)$. Position-dependent zeroth-order cross-coupling absent in flat space, encoding how the expanding geometry creates an effective charge density. Antisymmetric: no net monopole.

What $\gamma = 4\pi k^2$ means. The coupling constant is the winding number squared. It is k^2 because on the mass shell $\ell = k$: one factor of k comes from the metric (Axiom 4) and one from the Fourier mode (Corollary 7). Electromagnetic charge is topology.

Cone frame interpretation.

- **Gauge-diagonalised frame**: $g^{uv} = 0$. Purely real ODE. No coupling between a and b. Pure geometry. No electromagnetism.
- **Original frame**: $g^{uv} = -\pi k$. Complex ODE. a and b couple antisymmetrically. Maxwell's equations emerge.

The "two polarisations of light" are not fundamental. Light is one helical object $f(u)\, e^{i\ell v}$ propagating on the manifold. Electromagnetism is the projection of helical geometry onto a real basis.

Caveat. This derivation recovers the structural form of Maxwell's equations from the wave equation on the helix. The identification $a \leftrightarrow E$, $b \leftrightarrow B$ is a structural correspondence: it matches the algebraic signature (antisymmetric first-order cross-coupling) and recovers the correct flat-limit equations. Promoting this to a physical equivalence requires: (a) identifying the electromagnetic field strength $F_{\mu\nu}$ with a geometric quantity on the manifold; (b) recovering the Lorentz force law from geodesic deviation; and (c) matching the coupling constant $4\pi k^2$ to the observed electromagnetic coupling $e^2/(4\pi\epsilon_0)$. These are deferred to later work.

Verification. All coefficients C_2, C_1, C_0 computed symbolically via the Laplace–Beltrami operator in SymPy. Real/imaginary splitting confirmed. Antisymmetry of cross-coupling verified algebraically. Cone frame limit ($g^{uv} \to 0$) confirmed to annihilate all imaginary coefficients. Flat-limit Maxwell equations recovered and verified to produce the wave equation with phase velocity $c = 1$. Full computation in `maxwell_helix_full.py`.

Corollary 20 (Gravity from Chain Curvature). *To be promoted to revised standard. Sketch: $K = 0$ means a straight chain carries no gravitational field. Gravity = curvature of the chain centreline, sourced by phase energy $E = k^2/4$ per cell.*